

TrES-3b

R-2797

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130724 · created 2026-09-04T23:25:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456497.8602 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.055
Transit depth (Rp/Rs)^2	0.91 +/- 1.04 [%]
Orbital Inclination (inc)	77.23 +/- 2.85
Airmass coefficient 1 (a1)	1.43 +/- 0.16
Airmass coefficient 2 (a2)	-0.278 +/- 0.091
Scatter in the residuals of the lightcurve fit is	9.49 %
Best Comparison Star	#1 - [275, 133]
Optimal Aperture	2.45
Optimal Annulus	5.43
Transit Duration (day)	0.007 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.096 $\pm$ 0.055 vs 0.16309 (deviation 1.22 σ)
Duration fit vs published	0.007 d vs 0.0567 d (deviation 2.92 σ)
Mid-time O-C	-1.7 min
Depth SNR	1.7
Residual scatter	9.49 %

Light curve

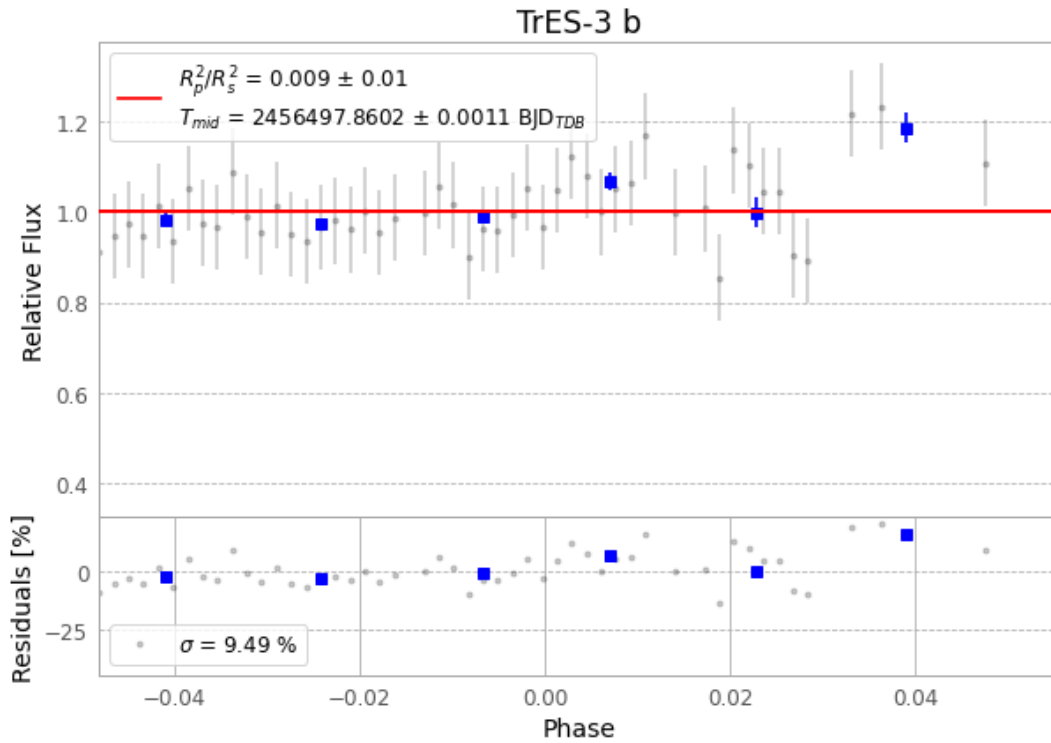


Figure 1 — FinalLightCurve_TrES-3 b_2013-07-24.png (SHA-256 e42424bfa0a66c18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6401ae66c3f40a3d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3130724070312.FITS → TRES-3130724101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130724.log (f72251e41337...), FinalLightCurve_TrES-3 b_2013-07-24.png (e42424bfa0a6...), FinalLightCurve_TrES-3 b_2013-07-24.csv (ebd0d6a15ffa...)

Compute resources

Wall time 775.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:25Z.